

PROJECT SCOPE

Organizational Brain + Legal Industry Module

Document status: Step 1 of Software Analysis & Design — SCOPE ONLY.

Traceability: This document is the authoritative source of truth for all subsequent requirements analysis, system analysis, system design, architecture, diagrams, project planning, and development work. Later design decisions must remain traceable to this scope.

1. Document Purpose

This document defines the boundary and intent of the system being built. It does not define requirements, actors, use cases, database schema, APIs, UML diagrams, architecture, or a project schedule — those are later steps. Its sole purpose is to answer, unambiguously: what is this system, what does V1 need to prove, and where does the line sit between what's in and what's out.

2. Product Definition

The system is an **Organizational Brain platform**: an intelligence layer that transforms an organization's scattered knowledge — documents, policies, decisions, historical cases, employee expertise, business-system data — into persistent, validated, reasoning-capable organizational intelligence.

The product is:

GENERAL ORGANIZATIONAL BRAIN + INDUSTRY-SPECIFIC MODULE

It is explicitly **not** a redefinition into any of the following, even though it may contain components that resemble them:

- a generic chatbot
- a CRM or ERP
- a document management system
- a vector database
- a RAG application
- a collection of independent, unconnected AI bots
- a simple knowledge repository

3. System Vision

An organization's most valuable knowledge is fragmented across people, documents, systems, and unrecorded experience, and is lost when an employee leaves or forgets. The Organizational Brain exists to make that knowledge organizational rather than personal: acquired, understood, validated, remembered, connected, retrievable, and reasoned over — so the organization gets progressively more intelligent rather than repeatedly relearning the same things.

4. System Purpose

The Brain exists to:

- acquire organizational knowledge from multiple sources
- understand and structure that knowledge
- validate it before treating it as trustworthy
- maintain it as persistent organizational memory
- retrieve what's relevant to a given question or situation
- reason over organizational context to produce grounded answers
- learn from new information, events, outcomes, and feedback
- surface contradictions or conflicts in what it has learned
- support (not replace) human decision-making with data-driven insight
- provide a shared foundation that specialized AI Employees and industry-specific modules can build on

5. General Organizational Brain

The General Brain is the reusable intelligent core, independent of any industry. At scope level (not technical-design level), it is responsible for:

- Organization and user management
- Organizational knowledge (acquisition, extraction, validation)
- Organizational memory
- Retrieval
- Reasoning
- Organizational events
- Learning
- Feedback
- Data Science / Machine Learning intelligence
- Decision intelligence
- AI Employee infrastructure
- Audit and governance
- Integration capabilities (connectors to external business systems)
- Organizational interaction / conversational interface

These are scope-level capability names only. Their internal design, schema, and APIs are explicitly deferred to later steps.

6. Legal Module

The Legal Module is the first industry-specific extension of the General Brain. It does not duplicate the Brain's capabilities — it specializes them:

GENERAL ORGANIZATIONAL BRAIN + LEGAL DOMAIN INTELLIGENCE = LEGAL ORGANIZATIONAL INTELLIGENCE SYSTEM

At scope level, the Legal Module may include:

- Legal knowledge (laws, regulations, internal legal policy)
- Legal documents
- Legal cases
- Legal entities
- Legal workflows
- Legal-specific rules
- Legal document analysis
- Case analysis
- Legal reasoning support
- A Legal AI Employee
- Legal-specific Data Science / ML use cases

The exact functional detail of each item above is deferred to requirements analysis (Step 2+). This document only fixes the **boundary**: the Legal Module consumes and specializes the General Brain; it does not contain a second, separate brain.

7. V1 Success Definition

V1 is **not** a production-scale system. V1 is a **functional, integrated, end-to-end proof of the vision** — a small but real working system that demonstrates the vision is technically and conceptually feasible, at prototype scale, through genuine (if simplified) behavior rather than simulated appearance.

V1 must demonstrate, at a meaningful prototype scale, each of the following:

#	Capability	What “demonstrated” means for V1
A	General organizational intelligence	The Brain's core mechanisms (knowledge, memory, retrieval, reasoning) are not hard-coded exclusively for legal data — described as a “general-purpose organizational intelligence layer,” never as AGI.
B	Legal domain intelligence	The Legal Module operates on top of the General Brain using a controlled set of laws/legal documents/legal knowledge, without reimplementing Brain mechanisms.
C	Knowledge acquisition	The Brain acquires knowledge from a limited but real set of sources (e.g. documents, employee-provided knowledge, one connected business system) — not every possible source.
D	Knowledge validation	A real distinction exists between Candidate Knowledge and Trusted Knowledge, with human validation in the loop.
E	Organizational memory	Validated knowledge becomes persistent memory, conceptually distinct from raw knowledge storage, and is used in retrieval/reasoning.
F	Retrieval and reasoning	A real path: Knowledge/Memory → Retrieval → Context → Reasoning → Answer, grounded in organizational information rather than a generic LLM response.
G	Learning	New validated information and feedback measurably change what the Brain retrieves/knows — a real mechanism, not a static “the AI learns” claim.
H	Contradiction / conflict detection	The Brain can detect and flag a potential conflict between two pieces of knowledge and surface it for human review, with source/authority/version context — concept-level correctness required, not perfect semantic conflict resolution.
I	External system integration	Real (if simplified/local) interaction with at least one external business system (e.g. a CRM) — real data flow, not a visual mock.
J	Data Science / ML	At least one genuine ML capability (prediction, classification, clustering, anomaly detection, forecasting, etc.) functioning as a distinct, cooperating capability alongside the LLM — the project is not reduced to LLM + RAG.
K	Decision intelligence	Knowledge + memory + events + ML predictions + business rules are combined into a decision context that supports, not replaces, human decision-making.
L	AI Employees	At least a limited demonstration that a specialized AI Employee operates on the shared General Brain rather than as an isolated bot.

8. V1 Vision Proof

V1 must prove that the following conceptual chain actually works, end to end, inside the running system:

```
Organization ↓  
Data / Knowledge / Experience ↓  
Knowledge Acquisition ↓  
Understanding ↓  
Candidate Knowledge ↓  
Human Validation ↓  
Trusted Knowledge ↓  
Organizational Memory ↓  
Retrieval ↓  
Reasoning ↓  
Answer / Insight / Decision Support ↓  
Outcome / Feedback ↓  
Learning ↓  
Updated Organizational Intelligence
```

How each step is implemented is deferred to later design steps. What's fixed here is that **every link in this chain must be real and observable**, not merely asserted.

9. System Boundary

Inside V1

- General Organizational Brain (knowledge, memory, retrieval, reasoning, learning concept, conflict detection)
- Legal Module (operating on top of the General Brain)
- Organizational knowledge lifecycle (acquisition → candidate → validated → trusted)
- Human validation of knowledge
- Organizational memory (persistent)
- Retrieval and reasoning (LLM-assisted, grounded in organizational data)
- Learning concept (real feedback-driven update mechanism, simplified allowed)
- Conflict / contradiction detection (concept-level, not perfect)
- At least one external business-system integration (real, simplified allowed)
- At least one Data Science / ML capability
- Decision-support capability
- Limited AI Employee capability (at least one, sharing the Brain)
- Organization / users / roles
- Audit / provenance concepts where required to support the above

Outside V1

- Large-scale production deployment
- Massive enterprise infrastructure
- Multiple industry modules beyond Legal
- Dozens of external integrations

- Fully autonomous, organization-wide learning
- Large-scale distributed architecture
- Perfect / comprehensive legal reasoning
- Unrestricted AGI framing of any kind
- Production-level optimization for many concurrent users
- Support for every possible organizational data source

10. In Scope vs Out of Scope

Area	In Scope	Out of Scope	Reason
Organizational Brain core	General knowledge/memory/retrieval/reasoning/learning/conflict-detection mechanisms	Fully autonomous, unsupervised organization-wide learning	V1 must prove the loop works with a human in it; full autonomy is a maturity step beyond a first proof
Industry modules	Legal Module only	Sales, Support, HR, Finance, Procurement, or any other module	Scope invariant: Legal is the only committed first module; others are future extensibility, not V1 deliverables
Knowledge sources	A limited, real set (e.g. documents, employee-provided knowledge, one connected business system)	Every possible organizational data source (email, WhatsApp, ERP, every CRM, etc.)	V1 needs to prove the mechanism, not achieve source-coverage completeness
Knowledge trust	Candidate vs Trusted distinction with human validation	Fully automated trust scoring with no human in the loop	Scope invariant: human validation remains important for V1
Memory	Persistent memory distinct from raw storage, usable in retrieval	Neuroscience-accurate memory modeling	The distinction is architectural/conceptual, not a literal cognitive-science implementation
Retrieval & reasoning	Grounded, sourced answers via retrieval + LLM reasoning	General-purpose ungrounded chatbot behavior	A generic LLM answer is explicitly not organizational intelligence
Conflict detection	Concept-level detection + human review surface	Fully automated, semantically perfect conflict resolution	V1 needs a real working mechanism, not a solved research problem
External integration	One real (simplified allowed) business-system connector, e.g. CRM	Many integrations, enterprise-grade connector library	One real integration is sufficient to prove the integration boundary works
Data Science / ML	At least one genuine ML capability, connected to the Brain	A full suite of production-tuned models	The project must not reduce to LLM+RAG, but V1 doesn't need an ML product line
Decision intelligence	Combined-context decision support surfaced to a human	Autonomous decision-making / replacing human authority	Scope invariant: the Brain supports decisions, it does not make them
AI Employees	At least one AI Employee sharing the General Brain	A full roster of AI Employees across departments	One demonstration is sufficient to prove the shared-Brain architectural concept
Deployment / infra	Local or small-scale runnable system	Production-scale, distributed, multi-region infrastructure	V1 is a feasibility proof, not a production launch
Audit / governance	Provenance concepts where needed to support validation/trust	Full enterprise governance/compliance program	Governance maturity is a post-V1 concern

11. V1 Limitations

There is a deliberate distinction between the **Vision** (the full Organizational Brain concept, described in Sections 2–6) and the **V1 Implementation Capability** (what actually gets built and demonstrated in this first version, per Sections 7–10).

V1 is allowed to use **simplified** implementations wherever appropriate, provided the architectural concept is preserved and the behavior is real.

- **Acceptable:** a small, local CRM integration that reads/writes real data.
- **Not acceptable:** a button labeled “CRM Connected” that performs no real interaction.
- **Acceptable:** a simplified conflict-detection rule that correctly flags a real contradiction in the demo data.
- **Not acceptable:** a “Conflicts” screen populated with hand-written fake conflict examples.
- **Acceptable:** one real, working ML model with honestly-reported (even modest) metrics.
- **Not acceptable:** a dashboard showing fabricated accuracy numbers.

Simplification of scale, polish, and coverage is expected in V1. Simulation of functionality that does not actually run is not.

12. Future Extensibility

The General Brain is intended to support future industry modules without requiring the Brain itself to be redesigned. Candidate future modules (explicitly **not** part of V1) include: Sales, Customer Support, HR, Finance, Procurement, and others.

Only Legal is the committed first industry module within the current project scope. Naming these future candidates here documents direction; it does not expand V1 scope to include them.

13. Scope Invariants

These principles must remain true throughout all future design and development work on this project:

- 1 The product is Organizational Brain + Industry Module.
- 2 Legal is the first industry module.
- 3 The General Brain is reusable across industries.
- 4 The Legal Module extends the Brain rather than replacing it.
- 5 Knowledge is not automatically trusted.
- 6 Memory is conceptually distinct from knowledge.
- 7 Retrieval is not the same thing as memory.
- 8 LLM output is not automatically organizational knowledge.
- 9 ML predictions are not automatically organizational knowledge.
- 10 Human validation remains important for trusted organizational knowledge.
- 11 Organizational events and outcomes can contribute to learning.
- 12 External systems are integrated through explicit boundaries (connectors), not embedded directly into the core.
- 13 Data Science is a real system capability, not decorative analytics.
- 14 AI Employees operate using the shared organizational intelligence layer — they are not independent isolated bots.
- 15 V1 is a functional proof of the vision, not a production-scale system.
- 16 Simplification is allowed; fake functionality is not.

14. Scope → Future Analysis Traceability

```
PROJECT SCOPE (this document) ↓  
REQUIREMENTS ANALYSIS ↓  
ACTORS & USE CASES ↓  
SYSTEM ANALYSIS ↓  
SYSTEM DESIGN ↓  
ARCHITECTURE ↓  
DATABASE / API / COMPONENT DESIGN ↓  
WBS / PERT / PROJECT PLANNING ↓  
RING PLANNING ↓  
IMPLEMENTATION ↓  
INTEGRATION ↓  
TESTING ↓  
V1 DEMONSTRATION
```

Later design decisions must remain traceable to this scope. Any requirement, actor, use case, architectural component, or ring that cannot be traced back to a capability named in Sections 5–10 above should be treated as scope creep and flagged for explicit re-approval before proceeding.

15. Scope Questions / Open Decisions

These are ambiguities identified while writing this scope. They are intentionally left open rather than silently resolved, and should be settled before or during Step 2 (Requirements Analysis):

- 1 **Which external business system will the V1 integration target?** The source materials name “a CRM” as an example, not a commitment. A specific system (or a well-defined mock/local CRM) needs to be chosen before requirements analysis can specify the connector.
- 2 **Which specific ML use case(s) will V1 implement?** The source materials list prediction, classification, clustering, anomaly detection, forecasting, case analysis, and recommendation as candidates for the Legal domain, without committing to one. This choice affects data requirements and should be made explicitly in Step 2, not assumed here.
- 3 **What is the demo organization / dataset for V1?** A concrete (if fictional) legal organization with representative documents, cases, and knowledge is implied by “a controlled set of legal documents” but not yet defined.
- 4 **What does “at least one AI Employee” concretely do for Legal?** The scope establishes that it must share the General Brain; its specific responsibilities are deferred to requirements analysis.
- 5 **What quantitative or qualitative bar defines “meaningful prototype scale”** for each V1 capability in Section 7 (e.g., how many documents, how many knowledge items, how many conflict examples)? This document intentionally leaves that undefined — it belongs to requirements/acceptance-criteria work, not scope.

None of these open items are answered here; they are flagged for Step 2.

16. Scope Closure Criteria

Step 1 (this document) is considered complete when all of the following are true:

- The product definition is unambiguous (Section 2).
- General Brain vs. Legal Module boundaries are clear (Sections 5–6).
- V1 success is explicitly defined (Section 7).
- In-Scope and Out-of-Scope boundaries are explicit (Sections 9–10).
- The system boundary is clear (Section 9).
- V1 limitations are documented, distinguishing Vision from V1 Implementation Capability (Section 11).
- Future extensibility direction is documented without expanding V1 scope (Section 12).
- Scope invariants are documented (Section 13).
- No technical design decisions (database, API, architecture, UML, WBS/PERT) are presented as final in this document.
- Open ambiguities are explicitly flagged rather than silently resolved (Section 15).
- This document can serve as the source of truth for Step 2.

Step 1 status: COMPLETE.